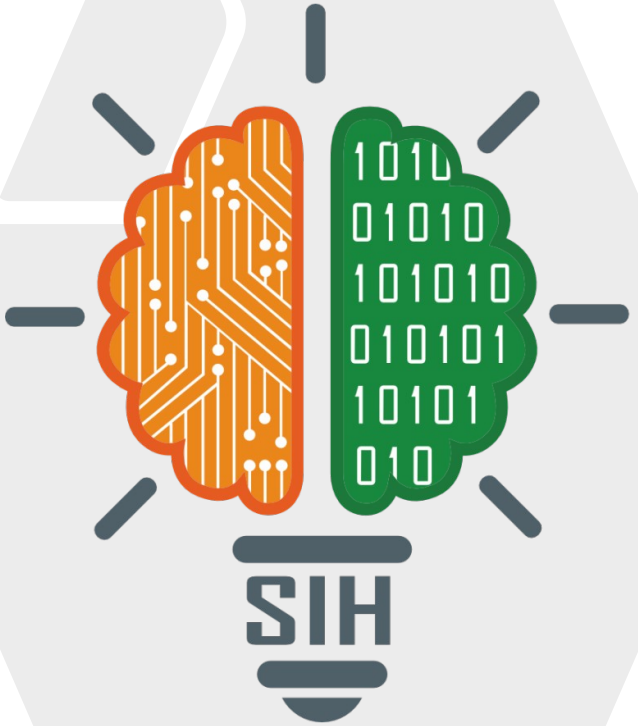
SMART INDIA HACKATHON 2026

SCOURGE

Ask the bank, not the paper.



- **Problem Statement ID –** SIH26265
- **Problem Statement Title –** Streamline of Security Deposit / FDR Portal
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** 199322
- **Team Name (Registered on portal) –** Arnavax#

SCOURGE

SCOURGE — ASK THE BANK, NOT THE PAPER

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Proposed Solution (Describe your Idea/Solution/Prototype)

The bank's own website proves the FDR: no bank API, no agreement, no paper. AICTE sees only that record.

Detailed explanation of the proposed solution

AICTE sends a **single-use challenge** for one FDR. The institution opens its own net banking; the connection key is **split with AICTE's verifier** (MPC-TLS), so nobody can fake the page. Login, OTP and PAN stay with the institution.

How it addresses the problem

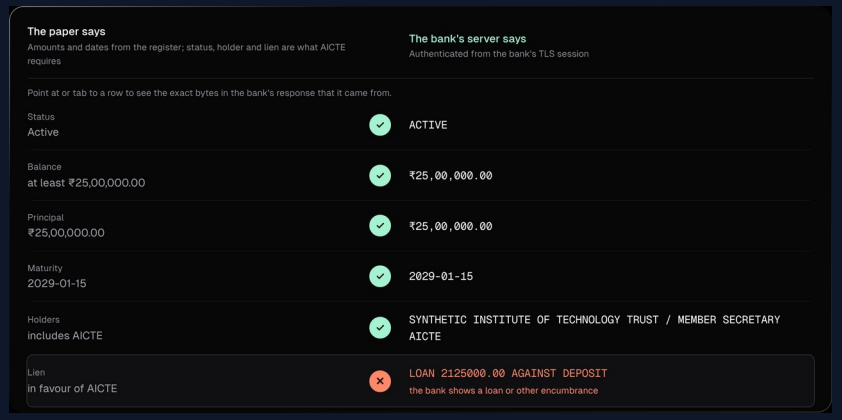
Paper is checked once and trusted for 10 years. Scourge **re-checks the bank any time and before every release**: a loan, early encashment, missing pledge (lien) or fake FDR shows up as a **Discrepancy**.

Innovation and uniqueness of the solution

Works with banks' existing net banking: no bank API or agreement needed, and API-ready when banks offer one. If a page cannot show a field: bank-signed e-advice, then branch confirmation. Fakes and replays are rejected.

LIVE PROTOTYPE · SYNTHETIC BANK

The bank's server contradicts the paper.



× HIDDEN LOAN CAUGHT

Every value traces to the bank's own bytes

The complete FDR lifecycle in one portal (all five stages run in the prototype)

1 FILE
Institution files online; renewals linked

→

2 PROVE
Bank-session proof on AICTE's challenge

→

3 MONITOR
T-90/60/30/7 alerts; adequacy; MT940/camt.053 reconcile

→

4 DECIDE
Verdict + evidence; hash-chained audit

→

5 PLEDGE RELEASE
Dept reviewers, 2 signers, holds; signed letter → bank ack

Why now: **₹201.42 crore of FDRs "do not appear in IDFC's records"** (CBI chargesheet, Oct 2026); AICTE joint FDRs found encashed or loaned against without consent.

SCOURGE

TECHNICAL APPROACH

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Technologies to be used

Proof layer
TLSNotary MPC-TLS · Rust
Selective disclosure · rustls

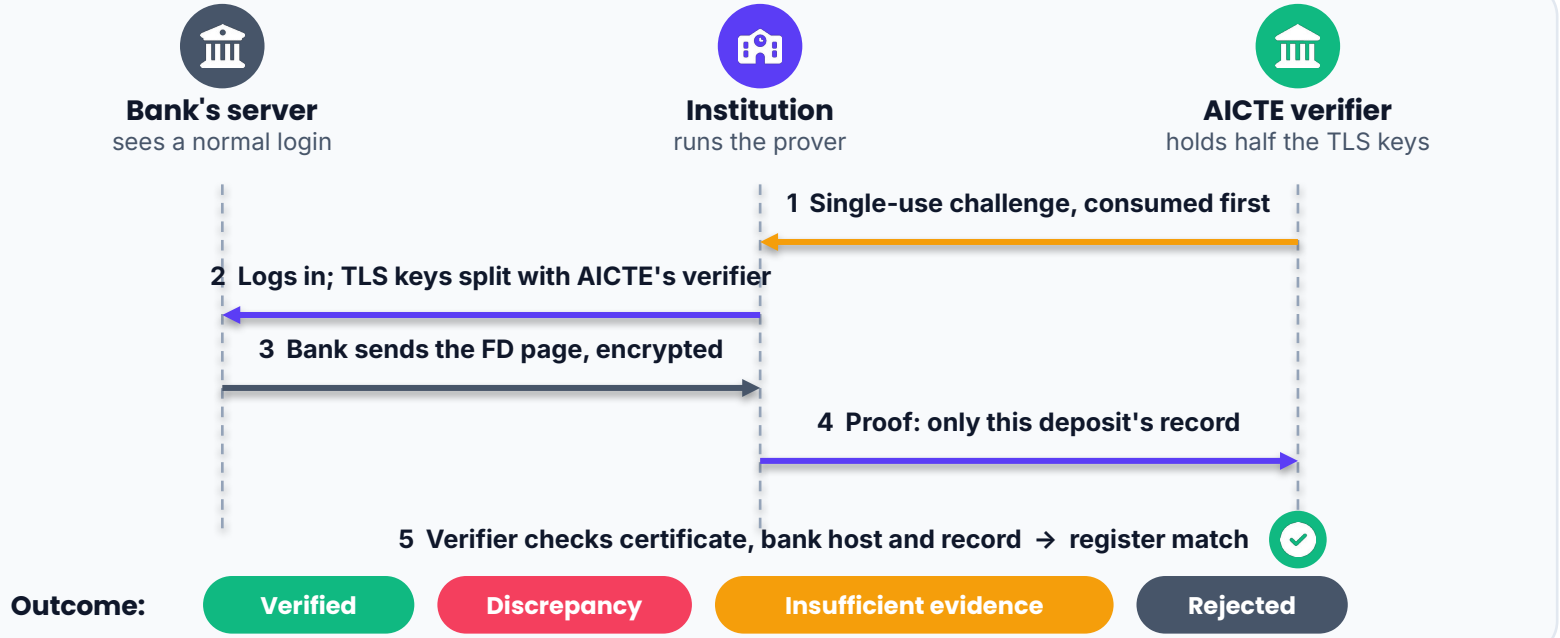
Backend
Python · FastAPI · SQLite
AES-GCM vault · hash-chained audit

Frontend
Server-rendered HTML · JS
Live 3-party proof view

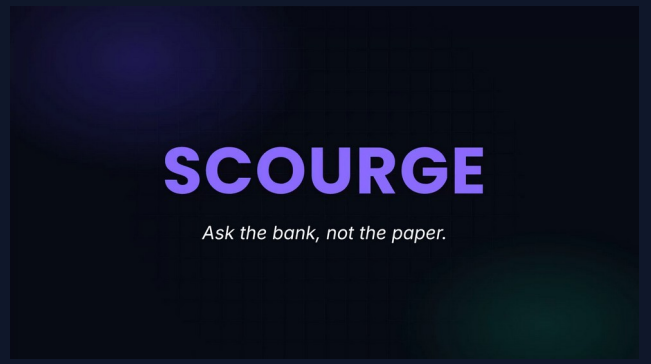
Ops & QA
Docker · nginx · GitHub CI
298 tests · 2FA · audits

Methodology and process for implementation (Flow Charts/Images/ working prototype)

In plain words: *an AICTE "notary" watches the bank page load, sees only the FD line, and nobody can change it.*



WATCH THE 3-MINUTE DEMO



Click to play here, or scan / click the QR for YouTube
Synthetic bank + live real-bank session

SCOURGE

FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.58 s
per bank proof
median of 60 runs · synthetic bank

Real bank
live split-key session
logged-in deposits API · 200 OK

298 / 298
automated tests pass
incl. rollback, forgery, backup-restore

₹0
software licences
60 MB/proof (measured) · FEASIBILITY.md

LIVE ON A REAL BANK
First bank tested: SBI · 1 Nov 2026

{"ok":true,"server":"api.onlineyono.sbi.bank.in","status_line":"HTTP/1.1 200 OK","secs":0.4}

Split-key session with a bank's logged-in deposits API (first bank tested: SBI); cert checked against public roots. Log: tools/realbank/RESULTS.md

Potential challenges and risks

Strategies for overcoming these challenges

Bank page can't be proven, or hides the lien	3-tier fallback: bank-signed e-advice, then branch; never a guess
Fake bank site or a replayed old proof	Rejected: bank certificate + single-use, 15-min challenge
Bank terms, consent and data protection	Own login, per-deposit consent, record-only disclosure; legal review
Bank encrypts data inside HTTPS, or ties it to the browser	Browser-extension prover; verifier gets the per-session app key

PILOT PLAN ✓ Prototype, 298 tests → ✓ Live real-bank session → FD pages at 3 large banks → 1 AICTE regional office → National rollout

SCOURGE

IMPACT AND BENEFITS

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Potential impact on the target audience

AICTE RB officers
A dated proof from the bank in seconds, not a letter in weeks

AICTE finance
The whole FDR book is re-checkable, before every release

Institutions
One short proof from their own net banking; Hindi or English

Students
29.6 lakh enrolled in 2025-26 (Lok Sabha reply); deposits checked all 10 years

Benefits of the solution (social, economic, environmental, etc.)

TODAY	WITH SCOURGE
Branch letter or uploaded screenshot	**The bank's own server, proven**
Checked once, trusted for 10 years	**Re-checked any time, before every release**
Weeks per confirmation	**~1 s per proof (synthetic, local); ~10 s at 50 Mbit/s**
Hidden loans found at release, or never	**Pledge checked at the bank; two-officer release**

TESTED ON 180 FRESH DEPOSITS
Random amounts, dates, formats · real proofs · synthetic bank

119 / 119	bad deposits the upload portal approved
0 / 119	bad deposits Scourge verified
0 / 61	honest deposits Scourge accused

Social
students' deposits stay intact

Economic
stops loss before release

Environmental
no paper letters or couriers

Governance
accounts, 2FA, vault, chained audit

SCOURGE

RESEARCH AND REFERENCES

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Details / Links of the reference and research work

The problem is real

CBI: ₹201.42 cr of FDRs not in IDFC's records (ThePrint, 7 Oct 2026)
theprint.in › cbi-says-haryana-pollution-board...

CBI on AICTE: joint FDRs encashed or loaned without consent
deccanherald.com › greater-transparency-aicte...

Regulation

AICTE guidelines 2024-27: cl. 6.2(g) 10-year release; cl. 11.8
indiankanoon.org/doc/173919027

Account Aggregator users must be regulated FIUs (DFS)
financialservices.gov.in › account-aggregator...

Technology

TLSNotary: MPC-TLS proofs of web data, MIT / Apache-2.0
tlsnotary.org

NeSL e-BG for security deposits: new instruments only (WDRA)
wdra.gov.in › ebg_requirement-27-08-2025

PROJECT RESOURCES

Demo video: youtu.be/Gm4dzjRQvC4
Live demo: scourge-sih26265.onrender.com · 298 tests · synthetic bank
Code + evidence: github.com/Lakki007/Arnavax-Scourge-SIH26265

Scourge vs existing approaches

	Upload & approve	Bank letter	Account Aggregator	NeSL e-BG	Scourge
Uses the bank's own data	✗	✓	✓	✓	✓
No bank API or agreement	✓	✓	✗	✗	✓
Institution cannot forge it	✗	~	✓	✓	✓
Covers existing FDR book	✓	✓	✓	✗	✓
Re-check in seconds	✗	✗	✓	✓	✓
Shown on a real bank today	✓	✓	✓	✓	✓ Live bank session

~ a bank letter can still be forged. Live split-key session with a real bank's logged-in deposits API: 200 OK (first bank tested: SBI); deposit fields not yet read.